

CBSE Class 10 Mathematics - Answer Key

1. The discriminant of the quadratic equation $3x^2 - 5x + 2 = 0$ is

Answer: (A)

Solution:

1. Identify the coefficients a , b , and c from the standard form $ax^2 + bx + c = 0$.
2. Here, $a = 3$, $b = -5$, and $c = 2$.
3. Use the discriminant formula $D = b^2 - 4ac$.
4. Substitute the values: $D = (-5)^2 - 4(3)(2)$.
5. Calculate the result: $D = 25 - 24 = 1$.

Derivation:

$$a = 3, b = -5, c = 2$$

$$D = b^2 - 4ac$$

$$D = (-5)^2 - 4(3)(2)$$

$$D = 25 - 24$$

$$D = 1$$

2. If $x = 3$ is a root of the quadratic equation $x^2 - 2kx - 3 = 0$, then the value of k is

Answer: (B)

Solution:

1. Since $x = 3$ is a root, it must satisfy the equation $x^2 - 2kx - 3 = 0$.
2. Substitute $x = 3$ into the equation: $(3)^2 - 2k(3) - 3 = 0$.
3. Simplify the expression: $9 - 6k - 3 = 0$.
4. Combine constant terms: $6 - 6k = 0$.
5. Solve for k : $6k = 6$, so $k = 1$.

Derivation:

$$(3)^2 - 2k(3) - 3 = 0$$

$$9 - 6k - 3 = 0$$

$$6 - 6k = 0$$

$$6k = 6$$

$$k = 1$$

3. Which of the following is a quadratic equation?

Answer: (C)

Solution:

1. A quadratic equation is of the form $ax^2 + bx + c = 0$ where $a \neq 0$.
2. Check (A): $x^2 + 3x = x^2 + 5$ simplifies to $3x - 5 = 0$ (Linear).
3. Check (B): $(x + 1)^2 = 2(x - 3)$ simplifies to $x^2 + 2x + 1 = 2x - 6$, which gives $x^2 + 7 = 0$ (Quadratic). Wait, checking options.
4. Check (C): $x^2 - 2x = (-2)(3 - x) = x^2 - 2x = -6 + 2x \Rightarrow x^2 - 4x + 6 = 0$ (Quadratic).
5. Check (D): $(x + 2)^3 = x^3 + 4$ simplifies to cubic equation.

Derivation:

$$(A)x^2 + 3x = x^2 + 5$$

$$\Rightarrow 3x - 5 = 0$$

$$(B)(x + 1)^2 = x^2 + 2x + 1$$

$$(C)x^2 - 2x = -6 + 2x$$

$$\Rightarrow x^2 - 4x + 6 = 0$$

$$(D)(x + 2)^3 = x^3 + 6x^2 + 12x + 8 = x^3 + 4$$

$$\Rightarrow 6x^2 + 12x + 4 = 0$$